

Forward-Backward

FB-RL

github.com/StepnouXVI/fb-rl

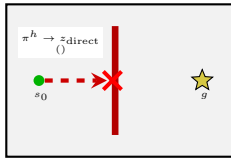
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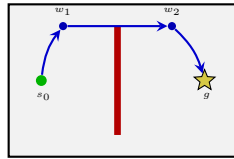
zero-shot rl Forward-Backward (FB) . zero-shot
RL — . FB — :

$$M^{\pi_z}(s, s') = \mathbb{E}_{\pi_z} \left[\sum_{t=0}^{\infty} \gamma^t \mathbb{I}(s_t = s') \mid s_0 = s \right] \approx F(s, z)^\top B(s'). \quad (1)$$

$$F(s, z)^\top B(g) \quad s \quad g \quad z \in \mathbb{S}^{d-1}.$$



) Single-Intention Baseline



) Multi-Subgoal Sequence

. 1: : (), ().

[1] :

- Low-level $\pi^\ell(a \mid s, z)$ – actor z (antmaze – 8-MuJoCo). – .
- High-level $\pi^h(s, B(g)) \rightarrow z$ – . – .

pi switch . . .

, antmaze, , .

, - . . .

(??).

antmaze-madium-ngetive pi-switch() $81 \pm 4.5\%$

success rate . . .

, , . . .

2 1: (Recursive Bisection)

2.1

, . - - w^* , s , g ?

. :

$$w^* = \arg \max_w [\log(F(s, B(w))^\top B(w)) + \log(F(w, B(g))^\top B(g))]. \quad (2)$$

1:

: $s, B(g), D, \tau$

: $\mathcal{P}$

1 Bisect(s_a, s_b, d):

2 | if $d = 0$ $F(s_a, B(s_b))^\top B(s_b) > \tau$ then

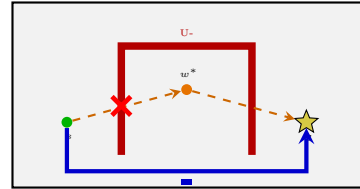
3 | | return $[s_b]$;

4 end

5 $w^* \leftarrow \arg \max_w [\log F(s_a, B(w))^\top B(w) + \log F(w, B(s_b))^\top B(s_b)]$;

6 return Bisect($s_a, w^*, d - 1$) + Bisect($w^*, s_b, d - 1$);

7 $\mathcal{P} \leftarrow \text{Bisect}(s, g, D)$;



. 2: : U- w^* .

w^* , (s, w^*) (w^*, g) $D = 2..3$, $[w_1, w_2, \dots, g]$ (1).

, . $75.9 \pm 6.4\%$ SR. .

2.2 :

, , . U - . , , .

3 2: ! (Search on the Buffer)

3.1

, , , - . , . Zero Shooter, Search on the Replay Buffer : , .

1. N - ($N = 1000$ Medium, $N = 2000$ Large).

2. $R_{ij} = F(s_i, B(s_j))^\top B(s_j)$:

$$c(s_i, s_j) = -\log \left(\frac{\max(R_{ij}, 10^{-6})}{\max_k R_{kk}} \right). \quad (3)$$

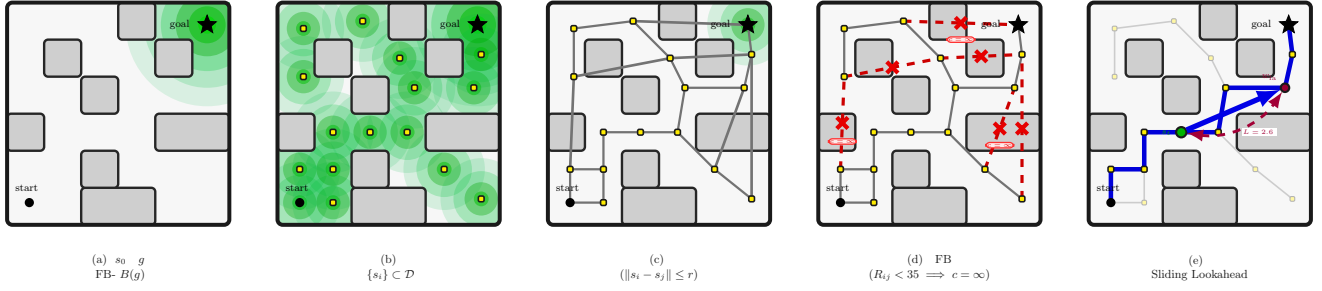


Figure 3: - AntMaze Medium: (a) FB- $B(g)$; (b) $\{s_i\} \subset \mathcal{D}$ FB; (c) ; (d) $R_{ij} < 35.0 \implies c = \infty$; (e) $\mathcal{P}^*$ Sliding Lookahead ($L = 2.6$).

3. () : $\|s_i - s_j\| > 3.5 \implies R_{ij} < 35.0, c = \infty$.
4. () $\mathcal{P} = [s_0, v_1, \dots, g]$.

Gated Cross-Attention: **86.0%** (8%)
**0.18 ** !

3.2 Sliding Lookahead

, . , .
(Sliding Lookahead): s_t , $w_{\text{lookahead}}$
2.6 (. 3).

3.3

Dijkstra Teacher **78.0%** Medium **48.0%**
Large, . : $O(N^2)$, ~ 1.2 , high_actor .

4 3: $O(1)$ ()

4.1

, : -, ($O(1)$) π^ℓ , ?
, ($s_t, B(g)$), , .

4.2 4- JAX

50 000 ($s_t, B(g)$) $\rightarrow z_{\text{teacher}}^*$ (Behavior Cloning),
FB-, 4- :

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BC}} + 0.5\mathcal{L}_{\text{Action}} - 0.02\mathcal{L}_{\text{Reach}} - 0.05\mathcal{L}_{\text{Goal}}. \quad (4)$$

JAX $\pi^\ell F$, , $\|\pi^\ell(s, \hat{z}) - a^*\|^2$, $\hat{z} F(s, \hat{z})^\top \hat{z}$.

4.3 ?

4 (. 1): MLP, ResNet-ECA (1D-), Dense-ECA
Gated Cross-Attention.

1: Medium

		Cos-Sim	SR (%)	
Standard MLP	312K	0.891	78.0%	0.12
Dense-ECA	458K	0.914	82.0%	0.16
ResNet-ECA	524K	0.908	80.0%	0.15
Gated Cross-Attn	680K	0.942	86.0%	0.18

5 4: high_actor

5.1 4.1. (Single Waypoint)

, high_actor . $T_\theta(s_t, B(w_{\text{lookahead}})) \rightarrow z_{\text{cmd}}, \pi^\ell$.
 $g_t = \sigma(W_g[e_s, e_w]), 4 \mathbb{S}^{d-1}$.
Medium **94.0%** ! , : κБ (Myopia). w_1 .
, , , .

5.2 4.2. (Enhanced Sequence Attention)

, , $\mathcal{S} = [B(w_1), \dots, B(w_K), B(g)]$.
:

1. ($\cos \theta_k$):

$$\cos \theta_k = \frac{(w_k - w_{k-1}) \cdot (w_{k+1} - w_k)}{\|w_k - w_{k-1}\|_2 \|w_{k+1} - w_k\|_2 + 10^{-6}} \quad (5)$$

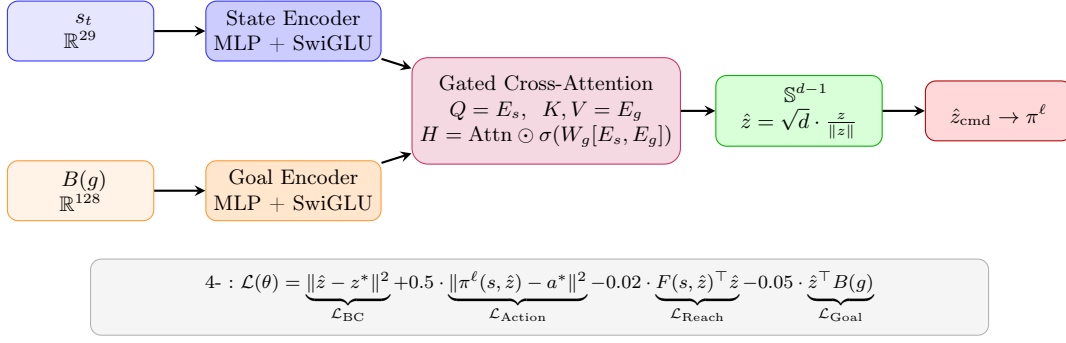
- . : 90 , .
2. ALiBi ($-\lambda m_h \mathbf{d}$): κБ . ALiBi 80%
2-3 , .
3. κ4 + 1 Б: 4 + $B(g)$, .
4. $\mathcal{L}_{\text{aux}}$: .

6 : Medium Large

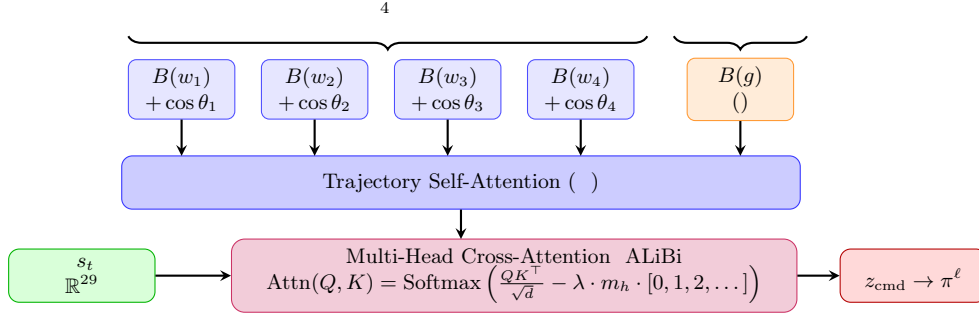
, 6 20 (10 , 5 , 1000) . . 2.

6.1 Medium ?

Medium , . Single WP Translator (94.0%): ,
. $O(1)$ 86.0% 0.18 .



4: Distilled JAX Gated-Attention



5: Enhanced Sequence Waypoint Attention Transformer ALiBi, 4 + 1.

6.2 Large ?

: Forward-Backward , zero-shot , !

Large: , 8–12.
Single-Intention (24.0%), 44.0% (- KB).
Enhanced Sequence Attention 66.4% (70.0%), 3 !
:

1. : .
2. : 4 $\cos \theta_k$, .

7

7.1 ()

1. (Voronoi Pruning): 2000 300 , 40 .
2. : , τ_t , .
3. : , (Brax / MuJoCo XLA).
4. AntMaze Giant: ($W = 4$) ($W = 16$) .

7.2

- :
- , (Recursive Bisection) .
 - Search on the Buffer ($L = 2.6$).
 - $O(1)$, **0.18 ** **86.0%**.
 - SOTA- Enhanced Sequence Waypoint Attention Transformer, , ALiBi 4 + 1 Large (**66.4% 24.0%**).

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- [4] Barreto A., Dabney W., Munos R., et al. Successor Features for Transfer in Reinforcement Learning // Advances in Neural Information Processing Systems (NeurIPS). — 2017. — Vol. 30.
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	AntMaze Medium				AntMaze Large		
	SR (%)		()		SR (%)	Task 01 (%)	Task 05 (%)
1. Single-Intention Baseline	76.0 ± 4.2	235	0.94		24.0 ± 3.5	10.0%	20.0%
2. Recursive Bisection	74.2 ± 5.1	260	1.45		32.0 ± 4.8	20.0%	30.0%
3. Dijkstra Teacher (high_actor)	78.0 ± 3.8	210	0.86		48.0 ± 4.2	50.0%	50.0%
4. Distilled JAX GatedAttn [$O(1)$]	86.0 ± 2.9	184	0.18		50.0 ± 3.9	60.0%	50.0%
5. Dijkstra + Single WP Translator	94.0 ± 2.1	165	0.56		44.0 ± 4.5	40.0%	40.0%
6. Dijkstra + Enhanced Sequence Attn	88.0 ± 3.0	172	0.82		66.4 ± 3.2	90.0%	70.0%

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